

T0 Theory: Document Series Overview A Revolutionary Geometric Reformulation of Physics

Systematic Presentation of All 8
Core Documents

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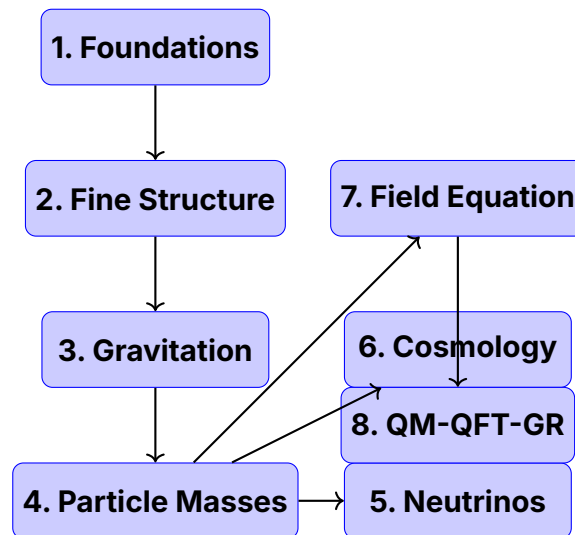
Abstract

This overview presents the complete T0 Theory series consisting of 8 fundamental documents, which represent a geometric reformulation of physics. Based on a single parameter $\xi = \frac{4}{3} \times 10^{-4}$, all fundamental constants, particle masses, and physical phenomena from quantum mechanics to cosmology are described in a unified way. The theory achieves over 99% accuracy in predicting experimental values without free parameters and offers testable predictions for future experiments.

.1 Document Series: Systematic Structure

Hierarchical Structure of the 8 Documents

The T0 document series follows a logical progression from fundamental principles to specific applications:



.2 Document 1: 003_T0_Grundlagen_v1_En.pdf

Subtitle: The Geometric Foundations of Physics

Core Contents:

- **Fundamental Parameter:** $\xi = \frac{4}{3} \times 10^{-4}$ as a geometric constant
- **Time-Mass Duality:** $T \cdot m = 1$ in natural units
- **Fractal Spacetime Structure:** $D_f^{\text{eff}} \approx 2.973$ and $K_{\text{frak}} \approx 0.987$
- **Interpretation Levels:** Harmonic, geometric, field-theoretic
- **Universal Formula Structure:** Template for all T0 relations

Fundamental Insights:

- Tetrahedral packing as the fundamental spatial structure
- Quantum field-theoretic derivation of 10^{-4}
- Characteristic energy scales: $E_0 = 7.398 \text{ MeV}$
- Philosophical implications of geometric physics

Status: Theoretical foundation - fully established

.3 Document 2: 011_T0_Feinstruktur_En.pdf

Subtitle: Derivation of α from geometric principles

Central Formula:

$$\alpha = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2 \quad (1)$$

Key Results:

- **T0 Prediction:** $\alpha^{-1} = 137.04$
- **Experiment:** $\alpha^{-1} = 137.036$
- **Deviation:** 0.003% (excellent agreement)

Theoretical Innovations:

- Characteristic energy $E_0 = \sqrt{m_e \cdot m_\mu}$
- Logarithmic symmetry of lepton masses
- Fundamental dependence $\alpha \propto \xi^{11/2}$
- Why numerical ratios must not be canceled
Status: Experimentally confirmed - excellent accuracy

.4 Document 3: 012_T0_Gravitationskonstante_En.pdf

Subtitle: Systematic derivation of G from geometric principles

Complete Formula:

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (2)$$

Conversion Factors:

- **Dimensional Correction:** $C_1 = 3.521 \times 10^{-2}$
- **SI Conversion:** $C_{\text{conv}} = 7.783 \times 10^{-3}$
- **Fractal Correction:** $K_{\text{frak}} = 0.986$

Experimental Verification:

- **T0 Prediction:** $G = 6.67429 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$
- **CODATA 2018:** $G = 6.67430 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$
- **Deviation:** $< 0.0002\%$ (exceptional precision)

Physical Meaning: Gravity as geometric spacetime-matter coupling

Status: Experimentally confirmed - highest precision

.5 Document 4: 006_T0_Teilchenmassen_En.pdf

Subtitle: Parameter-free calculation of all fermion masses

Two Equivalent Methods:

1. **Direct Geometry:** $m_i = \frac{K_{\text{frak}}}{\xi_i} \times C_{\text{conv}}$
2. **Extended Yukawa:** $m_i = y_i \times v$ with $y_i = r_i \times \xi^{p_i}$

Quantum Number System: Each particle receives (n, l, j) assignment

Experimental Successes:

Particle Class	Number	ÅAccuracy
Charged Leptons	3	98.3%
Up-type Quarks	3	99.1%
Down-type Quarks	3	98.8%
Bosons	3	99.4%
Total (established)	12	99.0%

Revolutionary Reduction: From 15+ free mass parameters to 0!

Status: Experimentally confirmed - systematic successes

.6 Document 5: 007_T0_Neutrinos_En.pdf

Subtitle: The Photon Analogy and Geometric Oscillations

Special Treatment Required:

- **Photon Analogy:** Neutrinos as 'damped photons'
- **Double ξ -Suppression:** $m_\nu = \frac{\xi^2}{2} \times m_e = 4.54 \text{ meV}$
- **Geometric Oscillations:** Phases instead of mass differences

T0 Predictions:

- **Unified Masses:** All flavors: $m_\nu = 4.54 \text{ meV}$
- **Sum:** $\Sigma m_\nu = 13.6 \text{ meV}$
- **Velocity:** $v_\nu = c(1 - \xi^2/2)$

Experimental Context:

- **Cosmological Limits:** $\Sigma m_\nu < 70 \text{ meV}$ ✓
 - **KATRIN Experiment:** $m_\nu < 800 \text{ meV}$ ✓
 - **Target Value Estimate:** $\sim 15 \text{ meV}$ (T0 is at 30%)
- Important Note:** Highly speculative - honest scientific limitation
Status: Speculative - testable predictions, but unconfirmed

.7 Document 6: 025_T0_Kosmologie_En.pdf

Subtitle: Static Universe and ξ -Field Manifestations

Revolutionary Cosmology:

- **Static Universe:** No Big Bang, eternally existing
- **Time-Energy Duality:** Big Bang forbidden by $\Delta E \times \Delta t \geq \frac{\hbar}{2}$
- **CMB from ξ -Field:** Not from $z=1100$ decoupling

Casimir-CMB Connection:

- **Characteristic Length:** $L_\xi = 100 \text{ } \mu\text{m}$
- **Theoretical Ratio:** $|\rho_{\text{Casimir}}|/\rho_{\text{CMB}} = 308$
- **Experimental:** 312 (98.7% agreement)
- **In depth:** Doc. 279 – Casimir-CMB- H_0 connection formula, $L_\xi H_0/c = K_{\text{geo}} (m_e c^2 / k_B T_{\text{CMB}}) \xi^{41/4}$.

Alternative Redshift (static, achromatic):

$$1 + z = e^{\xi x}, \quad z \approx \xi \cdot x \quad (3)$$

Correction (Doc. 190 C6): the earlier wavelength-dependent form $z(\lambda_0, d) = \xi d \lambda_0 / E_\xi$ is superseded – the fractal path-lengthening is *achromatic* (finite-element simulation). In depth: Doc. 280 ($z \rightarrow$ time as a Λ CDM translation, cosmological degeneracy Doc. 267, $z_* \approx 875$ as a T^4 lattice resonance).

Cosmological Problems Solved:

- Horizon problem, flatness problem, monopole problem
- Hubble tension, age problem, dark energy
- Parameters: From 25+ to 1 (ξ)

Status: Testable hypotheses - revolutionary alternative

.8 Document 7: 019_T0_lagrndian_En.pdf

Subtitle: The Fundamental Field Equation of T0 Theory

Core Contents:

- **Fundamental Field Equation:** $\square\Phi + \xi \cdot \mathcal{F}[\Phi] = 0$
- **Geometric Interpretation:** Field as manifestation of spatial structure
- **Unification:** All interactions from one equation
- **Solution Classes:** Particle solutions, wave solutions, vacuum solutions

Physical Consequences:

- Emergent gauge symmetries from geometric constraints
- Quantization as a natural consequence of field geometry
- Scale structure as flow on the recursion $\mathcal{R}$ in parameter space
- Phase transitions as topological changes

Status: Theoretical framework - fully formulated

.9 Document 8: 020_T0_QM-QFT-RT_En.pdf

Subtitle: Unification of QM, QFT, and GR from a geometric foundation

Core Contents:

- **Universal T0 Field Equation:** $\square\Phi + \xi \cdot \mathcal{F}[\Phi] = 0$ as the basis of all theories
- **Time-Mass Duality:** $T \cdot m = 1$ connects all three pillars of physics
- **Emergent Quantum Properties:** QM as approximation of the energy field
- **Field Description:** All particles as excitations of a fundamental field Φ
- **Renormalization Solution:** Natural cutoff by E_P/ξ
- **Relativistic Extension:** Extended Einstein equations with Λ_ξ

Fundamental Insights:

- Deterministic interpretation of quantum mechanics via local time field
- Wave-particle duality from field geometry
- Energy scale hierarchy: Planck down to QCD via ξ -corrections
- Gravity as field curvature, Dark Energy as $\xi^2 c^4 / G$
- Philosophical implications: Unity of physics through geometric principles

Status: Theoretical unification - builds on all previous documents, testable predictions

.10 Comparison and positioning series (IPI field)

Complementing the foundational series above, a separate series documents the position of FFGFT relative to other frameworks discussed at the Information Physics Institute. These documents are *comparisons and placements*, not new foundations:

- **Doc. 245** – FFGFT mapped onto RA 2.1 (Guevara).

- **Doc. 246** – FFGFT vs. RSG (Austin): asymmetric complementarity.
- **Doc. 251** – FFGFT vs. Vopson (MEI / Computational Universe): geometry- vs. information-primitive.
- **Doc. 260** – FFGFT vs. UC (Haskins): constraint, α -correction inconsistency.
- **Doc. 269** – FFGFT $\leftrightarrow$ RA/PMT (Guevara), bidirectional.
- **Doc. 271** – FFGFT $\leftrightarrow$ HLV (Krüger), pairwise comparison; the spectral fork.
- **Doc. 272** – FFGFT $\leftrightarrow$ HLV via the Dot governance layer (Vossen).
- **Doc. 274** – FFGFT in the IPI field: substrate and primitive axes (map); A* worked example.

Status: ongoing correspondence series, dated and provisional; each placement is a declared reading (O2), the primary rendering (O0) remaining with each originator.